



Statistical Techniques

Unit No 3: Correlation Analysis

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Syllabus

Unit III: Correlation Analysis

- Concept of Correlation
- Types of correlation
- Importance of correlation Analysis in Computer Application
- Measures of Correlation:
 - ❑ Karl Pearson's coefficient of Correlation
 - ❑ Spearman's Rank Correlation Coefficient.



Unit No 3: Correlation Analysis

Concept of Correlation



Meaning of Correlation

Correlation is a statistical measure that indicates the degree and direction of relationship between two or more variables. It shows how changes in one variable are associated with changes in another variable.

For example:

If coding practice increases and programming skills improve, these two variables are correlated.

Correlation is a statistical technique used to measure the degree and direction of relationship between two or more variables. It indicates how a change in one variable is associated with a change in another variable.



Correlation

However, correlation does not mean causation. Two variables may move together without one causing the other.

Example:

Ice cream sales and internet usage both increase in summer, but one does not cause the other.



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Types of Correlation



Types of Correlation

Correlation can be classified in three main ways:

1. Based on Direction

- Positive Correlation
- Negative Correlation
- Zero Correlation

2. Based on Number of Variables

- Simple Correlation
- Multiple Correlation
- Partial Correlation

3. Based on Nature/Form of Relationship

- Linear Correlation
- Non-Linear (Curvilinear) Correlation



Correlation based on direction :

1. Positive Correlation

Definition

Positive correlation occurs when both variables move in the same direction. ($r > 0$)

- **Increase in one variable → Increase in the other**

e.g. :

- Coding practice time increases → Coding efficiency increases
- When bandwidth increases, download speed increases.

So both variables move in the same direction, showing positive correlation.

- **Decrease in one variable → Decrease in the other**

e.g.:

- As RAM decreases, system performance also decreases.
- Database optimization decreases → Query speed decreases

Both variables move in the same direction, so it is positive correlation.



Correlation based on direction :

2. Negative Correlation

Definition

Negative correlation occurs when two variables move in opposite directions.

Negative correlation is a relationship between two variables in which an increase in one variable leads to a decrease in the other and vice versa.

Increase in one variable → Decrease in the other

Decrease in one variable → Increase in the other

The correlation coefficient is negative.

$r < 0$

e.g.

- As bugs decrease, software quality increases.
- When server load increases, system speed decreases.
- Network congestion increases, internet speed decreases.



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Correlation based on direction :

3. Zero Correlation

Definition

Zero correlation occurs when there is no relationship between two variables.

e.g.

1. Laptop price and student exam marks

Laptop price has no relation with marks.

2. Keyboard colour and programming skills

Keyboard Color has no relation with skill



Correlation Based on Number of Variables:

1. Simple Correlation

1. Simple Correlation

Simple correlation refers to the relationship between two variables only. In this type of correlation, we study how a change in one variable is associated with a change in another variable. It helps in understanding whether the two variables move together or in opposite directions. Simple correlation is the most basic form of correlation and is commonly used in statistical analysis, data science, and computer applications to analyze the relationship between two factors.

Examples

- Relationship between coding practice time and programming skill.
- Relationship between server RAM and system performance.
- Relationship between internet bandwidth and download speed.
- Relationship between number of bugs and software quality.



Correlation Based on Number of Variables:

2. Multiple Correlation

Multiple correlation refers to the relationship between one dependent variable and two or more independent variables. In this case, the effect of several variables on a single variable is studied simultaneously. This type of correlation is very useful in fields like machine learning, data analysis, and computer systems, where many factors influence a single outcome.

Examples

- Relationship between system performance and factors such as RAM and CPU speed.
- Relationship between website response time and server load and bandwidth.
- Relationship between database query speed and RAM and indexing techniques.
- Relationship between machine learning model accuracy and training data size and algorithm used.



Correlation Based on Number of Variables:

3. Partial Correlation

Partial correlation measures the relationship between two variables while keeping the effect of other variables constant. This means we analyze the connection between two variables after removing the influence of other related variables. Partial correlation is often used in advanced statistical analysis, artificial intelligence, and machine learning to understand the true relationship between variables.

Examples

- Relationship between study hours and exam marks while keeping intelligence level constant.
- Relationship between coding practice and programming performance while keeping study hours constant.
- Relationship between RAM and system performance while controlling CPU speed.
- Relationship between internet speed and download rate while keeping network congestion constant.



Based on Nature/Form of Relationship

1. Linear Correlation

Linear correlation refers to the relationship between two variables in which the change in one variable is associated with a proportional change in another variable, forming a straight-line pattern when plotted on a graph.

In this type of correlation, the rate of change between the variables remains constant. Linear correlation can be positive (both variables increase or decrease together) or negative (one variable increases while the other decreases). This type of correlation is widely used in statistical analysis, economics, and computer applications because it is simple to analyze and interpret.

Examples

- Relationship between study hours and exam marks.
- Relationship between coding practice time and programming skill.
- Relationship between internet bandwidth and download speed.
- Relationship between number of users on a website and server load.



Based on Nature/Form of Relationship

2. Non-Linear (Curvilinear) Correlation

2. Non-Linear (Curvilinear) Correlation

Non-linear or curvilinear correlation refers to the relationship between two variables where the change in one variable does not produce a constant change in the other variable. Instead of forming a straight line, the relationship forms a curve when represented graphically.

In such cases, the rate of change between the variables varies at different levels. Curvilinear correlation occurs in many real-life situations where the relationship between variables is complex and does not follow a straight-line pattern.

Examples

- Relationship between stress level and work performance (performance increases up to a point and then decreases).
- Relationship between experience and salary growth (rapid increase initially, slower growth later).
- Relationship between system performance and number of running applications (performance drops sharply after a certain limit).
- Relationship between study time and productivity (after excessive study, efficiency may decrease).
- Relationship between advertising expenditure and sales (sales increase initially but later stabilize).



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Importance of correlation Analysis in Computer Applications



Importance of correlation Analysis in Computer Applications

1. Understanding Relationship Between Variables

Correlation analysis helps determine how two variables are related.

Example:

- A software company studies the relationship between server load and response time. If server load increases and response time also increases, there is a positive correlation.

Use in Computer Applications

- Performance analysis of systems
- Network traffic monitoring
- Database performance evaluation



Importance of correlation Analysis in Computer Applications

2. Improving Software Performance

Developers use correlation to identify factors affecting application performance.

Example:

A programmer analyzes correlation between:

Number of users online

Application crash rate

If a strong correlation exists, developers can improve server scaling or load balancing.



Importance of correlation Analysis in Computer Applications

3. Data Science and Machine Learning

Correlation analysis helps select important features (variables) for machine learning models.

Example:

In a recommendation system (like Netflix or Amazon):

Correlation between user watch history and recommended movies helps build prediction models.

Use

Feature selection

Model accuracy improvement

Pattern discovery in datasets



Importance of correlation Analysis in Computer Applications

4. User Behavior Analysis

Web developers analyze correlation between user behavior metrics.

Example:

Correlation between:

Time spent on website

Number of purchases

If correlation is strong, companies improve UI/UX design to increase engagement.



Importance of correlation Analysis in Computer Applications

5. Cybersecurity and Fraud Detection

Correlation analysis helps detect suspicious patterns in system logs or network traffic.

Example:

Correlation between:

Multiple login attempts

Unauthorized access attempts

This helps in intrusion detection systems (IDS).



Importance of correlation Analysis in Computer Applications

6. Database and System Optimization

Correlation can help administrators optimize system resources.

Example:

Correlation between:

Database query size

Execution time

If queries with larger data show strong correlation with slower execution, optimization techniques like indexing or query restructuring can be applied.



Importance of correlation Analysis in Computer Applications

7. Predictive Analytics

Correlation helps predict future system behavior.

Example:

Correlation between:

Number of active users

Server CPU utilization

This helps companies predict server capacity requirements.



Correlation not only tells direction but also **how strong the relationship is.**

Correlation Value

Strength

± 1

Perfect

± 0.75

Strong

± 0.5

Moderate

± 0.25

Weak

0

No correlation

Unit No 3: Correlation Analysis

Measures of Correlation

- ❑ Karl Pearson's coefficient of Correlation
- ❑ Spearman's Rank Correlation Coefficient.



Measures of Correlation

Correlation measures the degree and direction of relationship between two variables.

To calculate correlation numerically, statisticians use different measures of correlation.

The two most commonly used measures are:

- **Karl Pearson's Coefficient of Correlation**
- **Spearman's Rank Correlation Coefficient**



1. Karl Pearson's Coefficient of Correlation

Karl Pearson's coefficient is the most widely used method to measure linear correlation between two quantitative variables.

It is also called Pearson's correlation coefficient and is denoted by r .

Example

Suppose a company studies the relationship between:

- Number of users on a server (X)
- Server response time (Y)

If the correlation coefficient is $r = 0.85$, it indicates a strong positive relationship, meaning response time increases as users increase.

Advantages

- Most accurate measure of linear correlation
- Uses all observations
- Widely used in data analysis and machine learning



Karl Pearson's Correlation Formula

$$r = \frac{\sum XY - \frac{(\sum X)(\sum Y)}{N}}{\sqrt{\left(\sum X^2 - \frac{(\sum X)^2}{N}\right) \left(\sum Y^2 - \frac{(\sum Y)^2}{N}\right)}}$$

Where:

- r = Karl Pearson's correlation coefficient
- X, Y = variables
- N = number of observations
- $\sum XY$ = sum of the product of X and Y
- $\sum X$ = sum of X values
- $\sum Y$ = sum of Y values
- $\sum X^2$ = sum of squares of X
- $\sum Y^2$ = sum of squares of Y



Range of r

$$-1 \leq r \leq +1$$

Value of r

Meaning

$$r = +1$$

Perfect positive correlation

$$r = -1$$

Perfect negative correlation

$$r = 0$$

No correlation



Example 1

A faculty member wants to study whether coding practice hours are related to programming test marks of MCA students. Data for five students is collected.

Student	Coding Practice Hours (X)	Programming Marks (Y)
A	12	52
B	18	60
C	24	68
D	30	76
E	36	84



Questions :

1. Calculate Coefficient of Correlation
2. Determine the type of correlation between the variables (positive, negative, or no correlation) and provide justification for your answer.



Solution

Student	X	Y	X^2	Y^2	XY
A	12	52	144	2704	624
B	18	60	324	3600	1080
C	24	68	576	4624	1632
D	30	76	900	5776	2280
E	36	84	1296	7056	3024



Solution

- $\Sigma X = 120$
- $\Sigma Y = 340$
- $\Sigma X^2 = 3240$
- $\Sigma Y^2 = 23760$
- $\Sigma XY = 8640$
- $N = 5$



Solution

$$r = \frac{8640 - \frac{(120)(340)}{5}}{\sqrt{(3240 - \frac{120^2}{5})(23760 - \frac{340^2}{5})}}$$

$$r = \frac{480}{480} = 1$$



Results

Ans Q. 1 : Coefficient of Correlation $(r) = +1$

Ans. Q. 2 Interpretation

There is a perfect positive correlation between coding practice hours and programming marks.

Justification

As coding practice hours increase, programming marks also increase consistently. Therefore, the variables move in the same direction, indicating strong positive correlation.



Example 2

A software company analysed the relationship between server capacity (in GB RAM) and the average website response time (in seconds) for its web application. The management collected the following data:

Server Capacity (GB)	Response Time (seconds)
20	90
30	80
40	70
50	60
60	50



Questions

Q. 1 Based on the above information, Calculate Coefficient of Correlation

Q. 2 If the company wants to further reduce the response time of the website, what decision should the management take regarding server capacity? Also explain whether investing in higher RAM servers would improve the overall performance of the website.



Solution

X	Y	X^2	Y^2	XY
20	90	400	8100	1800
30	80	900	6400	2400
40	70	1600	4900	2800
50	60	2500	3600	3000
60	50	3600	2500	3000



Totals:

- $\Sigma X = 200$
- $\Sigma Y = 350$
- $\Sigma X^2 = 9000$
- $\Sigma Y^2 = 25500$
- $\Sigma XY = 13000$
- $n = 5$



Numerator:

$$n\sum XY - \sum X \sum Y$$

$$= (5 \times 13000) - (200 \times 350)$$

$$= 65000 - 70000$$

$$= -5000$$

Denominator:

$$\sqrt{[(n\sum X^2 - (\sum X)^2)(n\sum Y^2 - (\sum Y)^2)]}$$

$$= \sqrt{[(5 \times 9000 - 200^2)(5 \times 25500 - 350^2)]}$$

$$= \sqrt{[(45000 - 40000)(127500 - 122500)]}$$

$$= \sqrt{[(5000)(5000)]}$$

$$= 5000$$

 **Correlation Coefficient**

$$r = -5000 / 5000$$

$$r = -1$$

Coefficient of Correlation $(r) = -1$

This indicates a perfect negative correlation between server capacity and response time.

Meaning: As server capacity increases, response time decreases perfectly.



Answer

Q. 2 If the company wants to further reduce the response time of the website, what decision should the management take regarding server capacity? Also explain whether investing in higher RAM servers would improve the overall performance of the website.

Answer :

The data shows a negative correlation between server capacity and response time. As server capacity (RAM) increases, the response time of the website decreases.

Therefore, if the company wants to further reduce the response time, the management should increase the server capacity by installing servers with higher RAM.

Investing in higher RAM servers will improve the overall performance of the website because more RAM allows the server to process multiple user requests faster, handle larger workloads efficiently, and reduce delays in loading web pages. This will lead to faster response time and better user experience.



Example 3

A software company analysed the relationship between hours spent on coding practice (per week) and number of bugs fixed in a project.

Coding Practice Hours (X)	Bugs Fixed (Y)
10	25
20	28
30	35
40	33
50	42



Questions

Q. 1 Calculate the coefficient of correlation between coding practice hours and bugs fixed.

Q. 2 Interpret whether increased coding practice always leads to higher bug fixing efficiency.

(Expected $r \approx 0.55-0.60$, moderate positive correlation)

Answer

Coding Practice Hours vs Bugs Fixed

X (Hours)	Y (Bugs Fixed)	X^2	Y^2	XY
10	25	100	625	250
20	28	400	784	560
30	35	900	1225	1050
40	33	1600	1089	1320
50	42	2500	1764	2100



Totals:

- $\Sigma X = 150$
- $\Sigma Y = 163$
- $\Sigma X^2 = 5500$
- $\Sigma Y^2 = 5487$
- $\Sigma XY = 5280$
- $n = 5$



Numerator

$$n\Sigma XY - \Sigma X \Sigma Y$$

$$= (5 \times 5280) - (150 \times 163)$$

$$= 26400 - 24450$$

$$= 1950$$

Denominator

$$\sqrt{[(n\Sigma X^2 - (\Sigma X)^2)(n\Sigma Y^2 - (\Sigma Y)^2)]}$$

First bracket:

$$5 \times 5500 - 150^2$$

$$= 27500 - 22500$$

$$= 5000$$

Second bracket:

$$5 \times 5487 - 163^2$$

$$= 27435 - 26569$$

$$= 866$$

So,

$$\sqrt{(5000 \times 866)}$$

$$\sqrt{4330000}$$

$$\approx 2081.6$$



Step 3: Correlation

$$r = 1950 / 2081.6$$

$$r \approx 0.94$$

Interpretation

- $r = 0.94$
- This shows a **strong positive correlation** between coding practice hours and bugs fixed.
- Developers who spend more time practicing coding tend to **fix more bugs efficiently**.



Example 4

A network administrator studied the relationship between **internet bandwidth (Mbps)** and **average download speed (MB/s)**.

Bandwidth (X)	Download Speed (Y)
20	18
30	22
40	25
50	24
60	29

Questions

1. Compute **Karl Pearson's coefficient of correlation**.
2. Comment whether increasing bandwidth always proportionally increases download speed.

(Expected $r \approx 0.56$)



Example 5

A database team analysed **database size (GB)** and **query execution time (seconds)**.

Database Size (X)	Query Time (Y)
10	12
20	18
30	22
40	21
50	27

Questions

1. Calculate the **correlation coefficient**.
2. Explain whether increasing database size affects query execution time.



Unit No 3: Correlation Analysis

Measures of Correlation

□ Spearman's Rank Correlation Coefficient.



Spearman's Rank Correlation

1. Meaning

Spearman's Rank Correlation is a non-parametric statistical method used to measure the relationship between two variables using ranks instead of actual numerical values.

It measures how well the relationship between two variables can be described using a monotonic function.

It is mainly used when:

- Data is in rank form
- Exact values are not available
- Data is qualitative or ordinal

Ranking of students in Programming and Database subjects.



Spearman's Rank Correlation

Example:

- Ranking of students in Programming and Database subjects.
- Ranking of Students in Programming and Data Structures
- Ranking of Databases by Query Speed and Reliability



Unit No 3: Correlation Analysis

Calculation of Spearman's Rank Correlation Coefficient. When Ranks are given



Formula

$$R = 1 - \frac{6 \sum d^2}{n(n^2 - 1)}$$



Example 1

A technology company ranked 6 software developers based on Coding Skill and Project Delivery Efficiency.

Developer	Coding Skill Rank	Delivery Rank
A	1	2
B	2	1
C	3	4
D	4	3
E	5	6
F	6	5



Questions

1. Calculate Spearman's Rank Correlation coefficient.
2. If the correlation is strong, should the company assign critical projects to developers with higher coding skill ranks?



Solution

Developer	Coding Rank (Rx)	Delivery Rank (Ry)	d = Rx – Ry	d ²
A	1	2	-1	1
B	2	1	1	1
C	3	4	-1	1
D	4	3	1	1
E	5	6	-1	1
F	6	5	1	1

$$\sum d^2 = 1 + 1 + 1 + 1 + 1 + 1 = 6$$



$$R = 1 - \frac{6 \sum d^2}{n(n^2 - 1)}$$

$$R = 1 - \frac{6(6)}{6(6^2 - 1)}$$

$$R = 1 - \frac{36}{6(36 - 1)}$$

$$R = 1 - \frac{36}{210}$$

$$R = 1 - 0.1714$$

$$R = 0.8286 \approx 0.83$$

Spearman's Rank Correlation Coefficient = 0.83



Answer 2

If the correlation is strong, should the company assign critical projects to developers with higher coding skill ranks?

Answer :

Yes.

Since the Spearman's Rank Correlation Coefficient ($R \approx 0.83$) shows a strong positive correlation between coding skill and project delivery efficiency, the company can reasonably conclude that developers with higher coding skill ranks tend to deliver projects more efficiently.

Therefore, the company should assign critical or complex projects to developers with higher coding skill ranks, as they are more likely to complete tasks accurately and on time.

However, management should also consider other factors such as experience, teamwork ability, and domain knowledge before making final project assignments. This will ensure better overall project performance.



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Example 2

A software company evaluated 8 programmers based on **Algorithm Design Skill** and **System Debugging Efficiency**. The ranks are given below.

Programmer	Algorithm Rank (R_x)	Debugging Rank (R_y)
A	1	3
B	2	1
C	3	4
D	4	2
E	5	7
F	6	5
G	7	8
H	8	6



Questions

Q. 1 Calculate Spearman's Rank Correlation .

Q. 2 Based on the calculated correlation, should the company assign complex debugging tasks to programmers with higher algorithm skill ranks? Justify your answer.

Q.3 If the company wants to improve debugging efficiency further, what managerial decision can be taken regarding training in algorithm design skills?



Answer

Q. 1 Spearman's Rank Correlation Coefficient = 0.762

This indicates a **strong positive correlation** between Algorithm Skill and Debugging Efficiency.

Q2. Decision Making

Yes, the company **should assign complex debugging tasks to programmers with higher algorithm skill ranks.**

Since the correlation coefficient is **0.762 (strong positive)**, it shows that programmers who are good at algorithm design also tend to perform better in debugging tasks. Therefore, assigning complex debugging work to such programmers will likely improve efficiency and accuracy.



Q3. Managerial Decision

To further improve debugging efficiency, the company should provide training programs focused on algorithm design and problem-solving techniques.

Better understanding of algorithms helps programmers:
analyze system errors more effectively

identify logical flaws quickly

improve overall debugging performance

Thus, investing in algorithm design training will enhance both coding quality and debugging efficiency in the organization.



Example 3

A software company evaluated **10 software engineers** based on their **Database Management Skill** and **System Optimization Performance**. The ranks assigned by the management are given below.

Engineer	Database Skill Rank (Rx)	Optimization Rank (Ry)
A	1	2
B	2	3
C	3	1
D	4	5
E	5	4
F	6	7
G	7	6
H	8	9
I	9	10
J	10	8

Questions

Q1. Calculate Spearman's Rank Correlation Coefficient between Database Skill and System Optimization

Q2. Interpret the value of the correlation coefficient and state whether the relationship between the two variables is positive or negative.

Q3. Based on the calculated correlation, should the company assign system optimization tasks to engineers with higher database skill ranks? Justify your answer.

Q4. If the company wants to improve overall system optimization performance, what managerial decision regarding database training should be taken?



$R=0.903$



Example 4

Two judges ranked 8 programmers in a coding competition.

Programmer	Judge A Rank	Judge B Rank
A	1	3
B	2	1
C	3	2
D	4	5
E	5	4
F	6	7
G	7	6
H	8	8



Find Spearman's Rank Correlation coefficient.



Solution

Programmer	A	B	d	d ²
A	1	3	-2	4
B	2	1	1	1
C	3	2	1	1
D	4	5	-1	1
E	5	4	1	1
F	6	7	-1	1
G	7	6	1	1
H	8	8	0	0



$$\Sigma d^2 = 10$$

$$n = 8$$

$$r_s = 1 - \frac{6 \Sigma d^2}{n(n^2 - 1)}$$

$$r_s = 1 - \frac{6(10)}{8(64 - 1)}$$

$$r_s = 1 - \frac{60}{504}$$

$$r_s = 0.881$$

Answer: Strong positive correlation.



Example 5

Two interview panels ranked 6 candidates for a software job.

Candidate	Panel A	Panel B
P	1	2
Q	2	3
R	3	1
S	4	5
T	5	4
U	6	6



Solution

Candidate	A	B	d	d ²
P	1	2	-1	1
Q	2	3	-1	1
R	3	1	2	4
S	4	5	-1	1
T	5	4	1	1
U	6	6	0	0



$$\sum d^2 = 8$$

$$n = 6$$

$$r_s = 1 - \frac{6(8)}{6(36 - 1)}$$

$$r_s = 1 - \frac{48}{210}$$

$$r_s = 0.771$$

Answer: High positive correlation.



Example 6

Two examiners ranked 7 MCA students in a project evaluation.

Student	Examiner A	Examiner B
A	1	2
B	2	1
C	3	4
D	4	3
E	5	6
F	6	5
G	7	7



Solution

Student	A	B	d	d ²
A	1	2	-1	1
B	2	1	1	1
C	3	4	-1	1
D	4	3	1	1
E	5	6	-1	1
F	6	5	1	1
G	7	7	0	0



A software company ranked **8 programmers** based on three skills:

Coding Skill (X)

Debugging Skill (Y)

Algorithm Design Skill (Z)

Programmer	Coding (X)	Debugging (Y)	Algorithm (Z)
A	1	2	3
B	2	1	2
C	3	3	1
D	4	5	4
E	5	4	5
F	6	6	6
G	7	7	8
H	8	8	7



Questions

Q. 1 Calculate Spearman's correlation between

Coding and Debugging

Coding and Algorithm Design

Debugging and Algorithm Design

Which two skills show the strongest relationship?

Q. 2 Based on the results, should the company assign complex algorithm development tasks to programmers with higher coding ranks?

Q. 3 If debugging and algorithm skills are strongly related, what training decision should management take



1. Correlation Between Coding (X) and Debugging (Y)

Student	X	Y	$d = X - Y$	d^2
A	1	2	-1	1
B	2	1	1	1
C	3	3	0	0
D	4	5	-1	1
E	5	4	1	1
F	6	6	0	0
G	7	7	0	0
H	8	8	0	0



$$\sum d^2 = 4$$

$$R = 1 - \frac{6(4)}{8(8^2 - 1)}$$

$$R = 1 - \frac{24}{504}$$

$$R \approx 0.95$$



2. Correlation Between Coding (X) and Algorithm (Z)

Student	X	Z	d	d ²	_____
A	1	3	-2	4	
B	2	2	0	0	
C	3	1	2	4	
D	4	4	0	0	
E	5	5	0	0	
F	6	6	0	0	
G	7	8	-1	1	
H	8	7	1	1	



$$\sum d^2 = 10$$

$$R = 1 - \frac{6(10)}{8(63)}$$

$$R = 1 - \frac{60}{504}$$

$$R \approx 0.88$$

3. Correlation Between Debugging (Y) and Algorithm (Z)

Student	Y	Z	d	d ²	_____
A	2	3	-1	1	
B	1	2	-1	1	
C	3	1	2	4	
D	5	4	1	1	
E	4	5	-1	1	
F	6	6	0	0	
G	7	8	-1	1	
H	8	7	1	1	



$$\sum d^2 = 10$$

$$R = 1 - \frac{60}{504}$$

$$R \approx 0.88$$

✓



Final Results

Correlation	Value
Coding – Debugging	0.95
Coding – Algorithm	0.88
Debugging – Algorithm	0.88

Interpretation

- **Strongest relationship:** Coding and Debugging (0.95)
- Programmers who are good at coding are usually good at debugging.
- Algorithm design is also strongly related but slightly less.



Q. 2 Based on the results, should the company assign complex algorithm development tasks to programmers with higher coding ranks?

Answer :

The company can assign debugging tasks to programmers with high coding ranks because both skills are strongly related.

Q. 3 If debugging and algorithm skills are strongly related, what training decision should management take

Answer :

Training programs can integrate coding and debugging practice together. Algorithm training should also be included but it shows slightly lower correlation compared to coding–debugging.



Unit No 3: Correlation Analysis

Calculation of Spearman's Rank Correlation Coefficient. When Ranks are not given



WHEN RANKS ARE NOT GIVEN

When we are given actual data and not the ranks we have to convert the data into ranks. We start by giving rank 1 to the highest (or lowest) value and give ranks accordingly to other values. But whether we start with lowest or highest value, we must follow the same method in case of all the observations.



Example 1

A software company evaluated 10 programmers based on their Algorithm Design Score and Debugging Efficiency Score during a coding competition.

Programmer	Algorithm Score (X)	Debugging Score (Y)
A	92	88
B	85	90
C	78	75
D	96	94
E	88	84
F	73	70
G	90	92
H	81	79
I	76	72



Questions:

Q1. Calculate Spearman's Rank Correlation Coefficient between Algorithm Design Score and Debugging Efficiency.

Q2. Based on the correlation result, should the company assign critical debugging tasks to programmers who have higher algorithm design scores? Justify your answer.

Q3. If the company wants to improve debugging performance, what training decision should management take regarding algorithm design skills?



Solution

Programmer	X	Rank R_x	Y	Rank R_y	$d = R_x - R_y$	d^2
A	92	2	88	4	-2	4
B	85	5	90	3	2	4
C	78	8	75	8	0	0
D	96	1	94	1	0	0
E	88	4	84	6	-2	4
F	73	10	70	10	0	0
G	90	3	92	2	1	1
H	81	7	79	7	0	0
I	76	9	72	9	0	0
J	84	6	86	5	1	1

$$R = 1 - \frac{6(14)}{10(10^2 - 1)}$$

$$R = 1 - \frac{84}{990}$$

$$R = 1 - 0.0848$$

$$R \approx 0.915$$

Interpretation

There is a very strong positive correlation between algorithm design score and debugging efficiency.



Q2. Based on the correlation result, should the company assign critical debugging tasks to programmers who have higher algorithm design scores? Justify your answer.

Answer :

Yes. Since correlation is strongly positive, programmers with higher algorithm design skills tend to perform better in debugging tasks. The company can assign complex debugging work to them.

Q3. If the company wants to improve debugging performance, what training decision should management take regarding algorithm design skills?

Answer :

Management should provide training in algorithm design and problem-solving techniques, because improving algorithm skills is likely to improve debugging efficiency.



Example 2

A software company analysed the relationship between Number of Coding Practice Hours per Week (X) and Software Testing Performance Score (Y) of 10 programmers.

A	45	78
B	38	92
C	52	70
D	60	88
E	41	75
F	55	95
G	47	72
H	36	80
I	50	85
J	43	68

Questions

Q1. Calculate Correlation coefficient

Q.2 Based on the calculated correlation, should the company assume that increasing coding practice hours will significantly improve software testing performance?

Q3. If the company wants to improve testing performance further, should management focus only on coding practice or also introduce specialized software testing training programs?



Prog	X	R_x	Y	R_y	d	d^2
A	45	6	78	6	0	0
B	38	9	92	2	7	49
C	52	3	70	9	-6	36
D	60	1	88	3	-2	4
E	41	8	75	7	1	1
F	55	2	95	1	1	1
G	47	5	72	8	-3	9
H	36	10	80	5	5	25
I	50	4	85	4	0	0
J	43	7	68	10	-3	9



$$R = 1 - \frac{6(134)}{10(10^2 - 1)}$$

$$\text{--- } R = 1 - \frac{804}{990}$$

$$\frac{804}{990} \approx 0.812$$

$$\text{--- } R \approx 0.19$$



Q.2 Based on the calculated correlation, should the company assume that increasing coding practice hours will significantly improve software testing performance?

Answer :

No. The calculated Spearman's Rank Correlation coefficient is $R \approx 0.19$, which indicates a weak positive correlation between coding practice hours and software testing performance. This means that although there is a slight positive relationship, increasing coding practice hours alone may not significantly improve testing performance. Other factors may also influence testing skills.



Q3. If the company wants to improve testing performance further, should management focus only on coding practice or also introduce specialized software testing training programs?

Answer :

Management should not rely only on coding practice hours. Since the correlation is weak, improving testing performance likely requires specialized software testing training programs, such as training in debugging techniques, automated testing tools, and quality assurance practices. Combining coding practice with focused testing training will help improve overall testing efficiency.



Example 3

A technology firm evaluates data analysts based on:

Python Programming Skill (X)

Data Visualization Skill (Y)

Machine Learning Model Accuracy (Z)

Analyst	Python Skill (X)	Visualization (Y)	ML Accuracy (Z)
A	85	80	88
B	70	68	72
C	92	90	95
D	65	60	63
E	88	84	90
F	60	62	58
G	78	75	80
H	95	92	97



Questions

Q. 1 Calculate Spearman's correlation between:

Python Skill & Visualization Skill

Python Skill & Machine Learning Accuracy

Visualization Skill & Machine Learning Accuracy

Decision-Making Questions

Q2. Which two technical skills are most strongly related?

Q3. If the company wants to design an advanced AI training program, should the focus be on Python programming or data visualization to improve machine learning performance?



Correlation Between Python Skill & Visualization

Analyst	Rx	Ry	d	d ²
A	4	4	0	0
B	6	6	0	0
C	2	2	0	0
D	7	8	-1	1
E	3	3	0	0
F	8	7	1	1
G	5	5	0	0
H	1	1	0	0



$$\sum d^2 = 2$$

$$r_{XY} = 1 - \frac{6(2)}{8(8^2 - 1)}$$

$$r_{XY} = 1 - \frac{12}{504}$$

$$r_{XY} \approx 0.976$$



Correlation Between Python Skill & ML Accuracy

Analyst	Rx	Rz	d	d ²
A	4	4	0	0
B	6	6	0	0
C	2	2	0	0
D	7	7	0	0
E	3	3	0	0
F	8	8	0	0
G	5	5	0	0
H	1	1	0	0



$$\sum d^2 = 0$$

$$r_{XZ} = 1$$



Correlation Between Visualization & ML Accuracy

Analyst	Ry	Rz	d	d ²
A	4	4	0	0
B	6	6	0	0
C	2	2	0	0
D	8	7	1	1
E	3	3	0	0
F	7	8	-1	1
G	5	5	0	0
H	1	1	0	0



$$\sum d^2 = 0$$

$$r_{XZ} = 1$$



Final Results

Correlation	Value
Python Skill & Visualization	0.98
Python Skill & ML Accuracy	1.00
Visualization & ML Accuracy	0.98



Q2. Which two technical skills are most strongly related?

The strongest relationship is between:

✓ Python Programming Skill and Machine Learning Accuracy

because the correlation coefficient is $r = 1.00$ (perfect positive correlation).

This means analysts with higher Python skills tend to achieve higher machine learning model accuracy.



Q3. Training Program Decision

If the company wants to improve Machine Learning performance, the training program should focus on:

✓ Python Programming

Reason:

Python Skill & ML Accuracy = 1.00 (highest correlation)

Visualization Skill & ML Accuracy = 0.98

Therefore, improving Python programming skills will have the greatest impact on machine learning model accuracy.



Unit No 3: Correlation Analysis

Calculation of Spearman's Rank Correlation Coefficient. For Repeated Ranks



Example 1

A multinational IT company wants to study the relationship between Employee Productivity Score (X) and Customer Satisfaction Rating (Y) for its project team members. The management collected data for 7 employees working on a major software development project. The scores are given below:

X	Y
78	70
85	75
78	72
90	80
78	75
85	68
92	82



Questions

Q. 1 Using Spearman's Rank Correlation Coefficient with tie correction, determine the relationship between employee productivity and customer satisfaction.

Q. 2 How should the software development manager use value of correlation coefficient while recruiting new programmers and designing training programs for developers?



Solution

Emp	X	Rx	Y	Ry	d = Rx - Ry	d ²
A	78	2	70	2	0	0
B	85	4.5	75	4.5	0	0
C	78	2	72	3	-1	1
D	90	6	80	6	0	0
E	78	2	75	4.5	-2.5	6.25
F	85	4.5	68	1	3.5	12.25
G	92	7	82	7	0	0


$$\sum d^2 = 19.5$$

Correction factor for x

- 78 repeated 3

$$T = \frac{3(9 - 1)}{12} = 2$$

- 85 repeated 2

$$T = \frac{2(4 - 1)}{12} = 0.5$$

$$T_x = 2.5$$

Correction Factor for Y

- 75 repeated 2

$$T = \frac{2(4 - 1)}{12} = 0.5$$

$$T_y = 0.5$$



$$R = 1 - \frac{6(\sum d^2 + \text{T.C.F.})}{n(n^2 - 1)}$$

$$R = 1 - \frac{6(19.5 + 2.5 + 0.5)}{7(49 - 1)}$$

$$R = 1 - \frac{6(22.5)}{336}$$

$$R = 1 - \frac{135}{336}$$

$$R = 1 - 0.4018$$

$$R \approx 0.60$$

Q. 1 Answer :

Final Answer

$R \approx 0.60$

Interpretation

The value 0.60 indicates a moderate positive correlation between:

Employee Productivity (X)

Customer Satisfaction Rating (Y)

This means that as employee productivity increases, customer satisfaction tends to increase as well, but the relationship is not perfectly strong. Other factors such as communication skills, teamwork, and problem-solving ability may also influence customer satisfaction.



Q. 2 How should the software development manager use value of correlation coefficient while recruiting new programmers and designing training programs for developers?

Answer :

How the Manager Should Use This Correlation Value

1. Recruitment of New Programmers

The manager should:

Prefer candidates with strong coding productivity and efficiency

Evaluate problem-solving ability and programming skills

Use coding tests, technical interviews, and project assignments during recruitment

Since productivity is positively related to customer satisfaction, hiring productive programmers can help improve client satisfaction levels.

2. Designing Training Programs

The manager should:

Conduct technical skill development programs

Provide training in efficient coding practices and software development tools

Introduce code review systems and mentoring programs

Encourage developers to improve debugging and problem-solving skills

Improving productivity through training can indirectly improve customer satisfaction.



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Example 2

A large software development company wants to examine the relationship between Coding Quality Score (X) and Project Manager Evaluation Score (Y) for developers working on an enterprise application project.

Developer	Coding Quality Score (X)	Project Manager Evaluation (Y)
A	70	65
B	75	68
C	75	70
D	80	72
E	85	75
F	85	75
G	90	80



Questions

Q. 1 Calculate Spearman's Rank Correlation Coefficient between Coding Quality Score (X) and Project Manager Evaluation Score (Y).

Q. 2 Based on the obtained value of Spearman's rank correlation, explain the relationship between the variables and recommend actions for the company



Answer

X	Rank	Y	Rank
70	1	65	1
75	2.5	68	2
75	2.5	70	3
80	4	72	4
85	5.5	75	5.5
85	5.5	75	5.5
90	7	80	7



Dev	Rx	Ry	d = Rx - Ry	d ²
A	1	1	0	0
B	2.5	2	0.5	0.25
C	2.5	3	-0.5	0.25
D	4	4	0	0
E	5.5	5.5	0	0
F	5.5	5.5	0	0
G	7	7	0	0

$$\sum d^2 = 0.5$$

Calculation of Correction Factor

Ties in X (Rx):

- 75 repeated 2 times
- 85 repeated 2 times

Formula:

$$T = \frac{m(m^2 - 1)}{12}$$

For 75 ($m = 2$)

$$T = \frac{2(4 - 1)}{12} = \frac{6}{12} = 0.5$$

For 85 ($m = 2$)

$$T = 0.5$$

$$T_x = 1$$

Tie Correction in Y

Tie in Y:

- 75 repeated 2 times

$$T = \frac{m(m^2 - 1)}{12} \quad (m = 2)$$

$$T_y = \frac{2(4 - 1)}{12}$$

$$T_y = 0.5$$

Total Correction Factor

$$\text{T.C.F.} = T_x + T_y = 1 + 0.5 = 1.5$$



Spearman Rank Formula with Ties

$$R = 1 - \frac{6(\sum d^2 + \mathbf{T.C.F.})}{n(n^2 - 1)}$$

Substitute values:

$$R = 1 - \frac{6(0.5 + 1.5)}{7(7^2 - 1)}$$

$$R = 1 - \frac{6(2)}{7(48)}$$

$$R = 1 - \frac{12}{336}$$

$$R = 1 - 0.0357$$

$$R \approx 0.964$$



Q.1 : Answer

Final Spearman Rank Correlation

$R \approx 0.96$

Interpretation

There is a **very strong positive correlation** between the two variables.



Q. 2 Based on the obtained value of Spearman's rank correlation, explain the relationship between the variables and recommend actions for the company

Answer :

The value of $R=0.96$ indicates a very strong positive correlation between the two variables: Coding Quality Score and Project Manager Evaluation Score

This means that as the coding quality of developers increases, their evaluation scores from project managers also tend to increase.

Recommendation for the Company

- **Focus on Code Quality Improvement**

The company should invest in training programs, coding standards, and code review practices to help developers improve their coding quality.

- **Encourage Best Practices in Development**

Implement peer reviews, automated testing, and clean coding guidelines to maintain high-quality software development.

- **Use Coding Quality as a Performance Indicator**

Since coding quality is strongly associated with manager evaluations, the company can include coding quality metrics in performance appraisal systems.

- **Provide Skill Development Opportunities**

Organize workshops, mentorship programs, and advanced programming courses to enhance developer skills.



Example 3

A software company studies relationship between:

Debugging Efficiency (X)

Client Satisfaction Score (Y)

Employee	X	Y
A	60	55
B	65	60
C	65	60
D	70	70
E	75	75
F	75	75
G	80	75



Questions

Q. 1 Calculate Spearman's Rank Correlation Coefficient

Q. 2 Based on the obtained value of Spearman's rank correlation, explain the relationship between the variables and recommend actions for the company



Solution

Employee	R _x	R _y	d	d ²
A	1	1	0	0
B	2.5	2.5	0	0
C	2.5	2.5	0	0
D	4	4	0	0
E	5.5	6	-0.5	0.25
F	5.5	6	-0.5	0.25
G	7	6	1	1



For X: two ties of 2 values each

$$T_x = 2 \times \frac{(2^3 - 2)}{12} = 2 \times \frac{6}{12} = 1$$

For Y: one tie of 2 and one tie of 3

$$T_y = \frac{6}{12} + \frac{24}{12} = 0.5 + 2 = 2.5$$



Solution

$$\begin{aligned} R &= 1 - \frac{6(\sum d^2 + T_x + T_y)}{n(n^2 - 1)} \\ &= 1 - \frac{6(1.5 + 1 + 2.5)}{7(49 - 1)} \\ &= 1 - \frac{6 \times 5}{7 \times 48} = 1 - \frac{30}{336} = 1 - 0.0893 = 0.91 \text{ (approx)} \end{aligned}$$



Q. 2 Based on the obtained value of Spearman's rank correlation, explain the relationship between the variables and recommend actions for the company

Answer :

There is a **very strong positive correlation** between debugging efficiency and client satisfaction.

This means:

Better coding/debugging → Higher client satisfaction

Developers who perform well technically also perform well in client evaluation

Recommendations for Company

Focus on Technical Skill Development

- Conduct debugging & coding optimization training

- Use real-world problem-solving sessions

Use Debugging Efficiency as a KPI

- While evaluating employees, include debugging performance as a major metric

Project Allocation Strategy

- Assign high-performing developers to **client-critical projects**

Mentorship Programs

- High scorers (like E, F, G) can mentor low scorers

Performance Appraisal

- Link promotions and incentives with both coding quality and client satisfaction



Example 4

A company studies the relationship between:
Number of Bugs (X) and **Client Satisfaction Score (Y)**

Employee	X (Bugs ↑)	Y (Satisfaction ↓)
A	12	90
B	18	85
C	18	85
D	25	80
E	30	75
F	30	70
G	40	70

↓)



Questions

Q1.

Calculate Spearman's Rank Correlation Coefficient (Answer : ≈ -0.9 (**strong negative**))

Q2.

Based on the result, explain how increasing bugs affects client satisfaction and suggest corrective measures.



Example 5

A software firm studies:

Time Taken to Resolve Issues (X) and **Customer Rating (Y)**

Employee	X (Time t)	Y (Rating t)
A	15	95
B	20	90
C	20	90
D	28	85
E	35	80
F	35	75
G	45	75



Questions

Q1.

Compute Spearman's Rank Correlation Coefficient.

(Answer : -1 (very strong negative))

Q2.

Suggest how the company can reduce resolution time while improving customer ratings.



Thank You



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